

UMC 0.11um Logic and Mixed-Mode Metal- Metal Capacitor Al Advanced Enhancement Resistor SPICE Model Document

-Monte Carlo-

**Version 0.6 Phase 1
2016/07/19**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.2_p1	2010/07/21	Ho Cheng Hu	1. Original release. 2. This Monte Carlo model including both process and mis-matching characteristics is extracted from L110AE_RES_V021.mdl L110AE_RES_V021.lib
0.3_p1	2011/08/03	Ho Cheng Hu	1. Update mismatch value - rsnd : from 1.41e-2 to 1.37e-2 - rspd : from 1.49e-2 to 1.52e-2 - rsnpo : from 1.12e-2 to 1.13e-2 - rsppo : from 1.15e-2 to 1.13e-2 2. Correct equation error in SPECTRE model card 3. This Monte Carlo model including both process and mis-matching characteristics is extracted from L110AE_RES_V021.mdl L110AE_RES_V021.lib
0.5_p1	2015/03/23	Cheng Tse Chou	1. Add model: r_m1_l110e r_1x_l110e r_2x_l110e 2. This Monte Carlo model including both process and mis-matching characteristics is extracted from l110ae_res_v051.mdl l110ae_res_v051.lib
0.6_p1	2016/07/19	Tzu Kang Hsiung	1. . Add the Monte Carlo plots of rnmwell_mml110e into the document 2.This Monte Carlo model including both process and mis-matching characteristics is extracted from

			l110ae_res_v061.mdl l110ae_res_v061.lib
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1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

0.11 um Logic and Mixed-Mode Metal Metal Capacitor Al Advanced
Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

1.3 Model Usage

1.3.1 Simulators

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1%. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9-SP2
Cadence SPECTRE version 15.1.0.257 64bit
Mentor Graphics ELDO version 15.4

1.3.2 The combination of statistical model & corner model

The Monte-Carlo model and corner model (3 corners) are combined into a single model library. User can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

mc: Monte-Carlo model
tt : Typical resistor case
ss : Maximum resistor case
ff : Minimum resistor case

1.3.3 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, user can run Monte-Carlo mismatching simulation over a corner model. For example, user can evaluate the mismatching performance at the tt case.

1.3.4 Sigma number for Monte-Carlo model

User can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

1.3.5 Model usage

The definitions of sub-circuit model parameters

After invoking Monte-Carlo model parameters, the compact resistor models evolve to sub-circuit models. The input parameters for the sub-circuit model are the same as the compact model, except input parameter (mis_flag) are added for the sub-circuit model.

The following is the parameter list:

mis_flag: the flag parameter is used to turn on/off mismatching parameters for individual devices. 1 is turn-on and 0 is turn-off. Default is 1.
w: resistor width
l: resistor length
m: number of devices
mf: the same as "m"

Follow these steps to invoke the model file during simulation.

- Define the flag of Monte-Carlo simulation.

 - PROCESS: 1 to turn on the MC process simulation
0 to turn off the MC process simulation

 - Example: .param process=1
.param process=0

- Define sigma value for MC simulation.

 - Example: .param sigma=3

- Define the number of Monte Carlo iterations.

 - Example: sweep monte=500 (for HSPICE and ELDO)
montecarlo numruns=500 (for SPECTRE)

For Synopsys HSPICE

- Copy model files into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
.option scale=0.9  
.lib "/ModelFileName_mc_corner.lib" CaseName  
, where CaseName could be mc, tt, ss, ff.
```

- Define device condition by "Xxx" statement.

```
Example: X1 n1 n2 pw DeviceName w=1.00E-06 l=1.00E-05 mf=1  
mis_flag=1
```

For Cadence SPECTRE

- Copy model files into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
SetOption1 options scale=0.9  
include "/ModelFileName_mc_corner.lib.scs" section=CaseName  
, where CaseName could be mc, tt, ss, ff.
```

- Define device condition by "Xxx" statement.

```
Example: X1 (n1 n2 pw) DeviceName w=1.00E-06 l=1.00E-05 mf=1  
mis_flag=1
```

For Mentor Graphics ELDO

- Copy following files into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.

```
.option scale=0.9  
.lib "/ModelFileName_mc_corner.lib.eldo" CaseName  
, where CaseName could be mc, tt, ss, ff.
```

- Define device condition by "Xxx" statement.

```
Example: X1 n1 n2 pw DeviceName w=1.00E-06 l=1.00E-05 mf=1  
mis_flag=1
```

ModelFileName: l110ae_res_v061_mc

The available DeviceName in this model are:

Non-salicide N+ diffusion resistor: rnnd_l110e
Non-salicide P+ diffusion resistor: rnpd_l110e
Non-salicide N+ poly resistor: rnnpo_l110e
Non-salicide P+ poly resistor: rnppo_l110e
Non-salicide HR poly resistor: rnhr1000_l110e
Salicide N+ diffusion resistor: rsnd_mml110e
Salicide P+ diffusion resistor: rspd_mml110e

Salicide N+ poly resistor: rsnpo_mml110e
Salicide P+ poly resistor: rsppo_mml110e
NWell resistor : rnwell_mml110e
M1 resistor : r_m1_l110e
1x resistor : r_1x_l110e
2x with 8k metal resistor : r_2x_l110e

This model is valid within the following geometry, voltage and temperature ranges.
Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

Geometry range:(before 90% shrinkage)

- a. Non-salicide N+ Poly resistor : rnnpo_l110e
 $1\ \mu\text{m} < \text{LDES} < 18\ \mu\text{m}$
 $0.5\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$
- b. Non-salicide P+ Poly resistor : rnppo_l110e
 $1\ \mu\text{m} < \text{LDES} < 18\ \mu\text{m}$
 $0.5\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$
- c. Non-salicide N+ Diffusion resistor : rnnd_l110e
 $1\ \mu\text{m} < \text{LDES} < 18\ \mu\text{m}$
 $0.5\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$
- d. Non-salicide P+ Diffusion resistor : rnpd_l110e
 $1\ \mu\text{m} < \text{LDES} < 18\ \mu\text{m}$
 $0.5\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$
- e. Non-salicide HR Poly resistor : rnhr1000_l110e
 $1\ \mu\text{m} < \text{LDES} < 18\ \mu\text{m}$
 $0.5\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$
- f. NWell resistor : rnwell_mml110e
 $1.67\ \mu\text{m} < \text{WDES} \leq 20\ \mu\text{m}$

Temperature range:

-55 °C ~ +135 °C

2 Monte Carlo DC Simulation Results for Process Variation

Five hundred (500) Monte Carlo iteration performed in the following simulation conditions, and the following corner points show tt, ff and ss values, individually

A reasonable number of Monte Carlo iteration is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)^{-1/2}.

2.1 Simulation Result Plots

The following figures show the Monte Carlo simulation results of Resistor (Before shrink : W=1um,L=5um) at 25 °C .

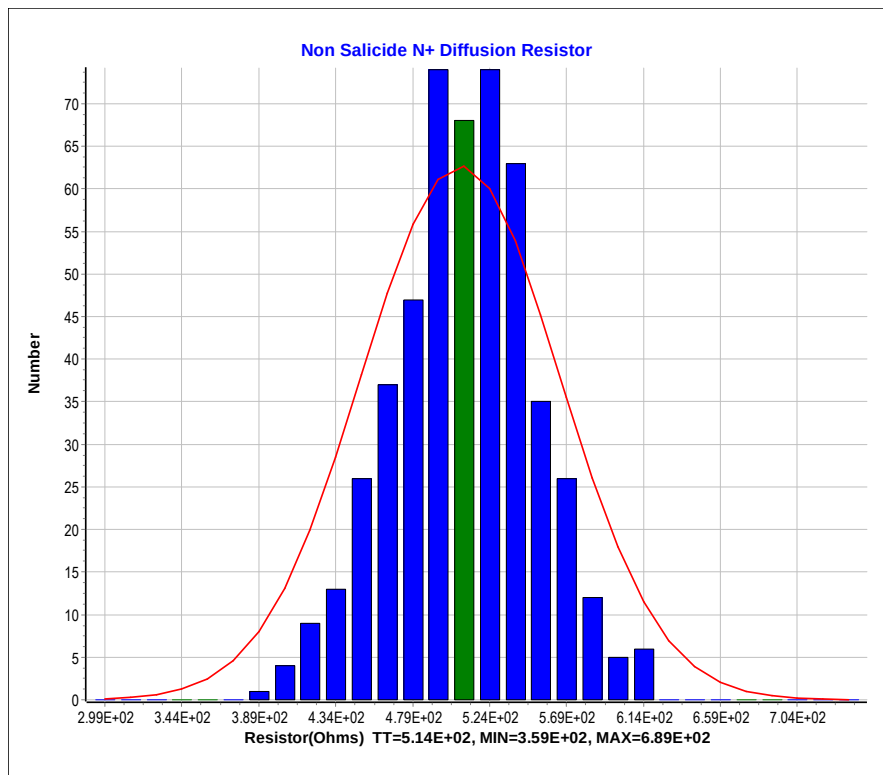


Figure 2-1 Monte Carlo plots of rnnd_l110e Resistor

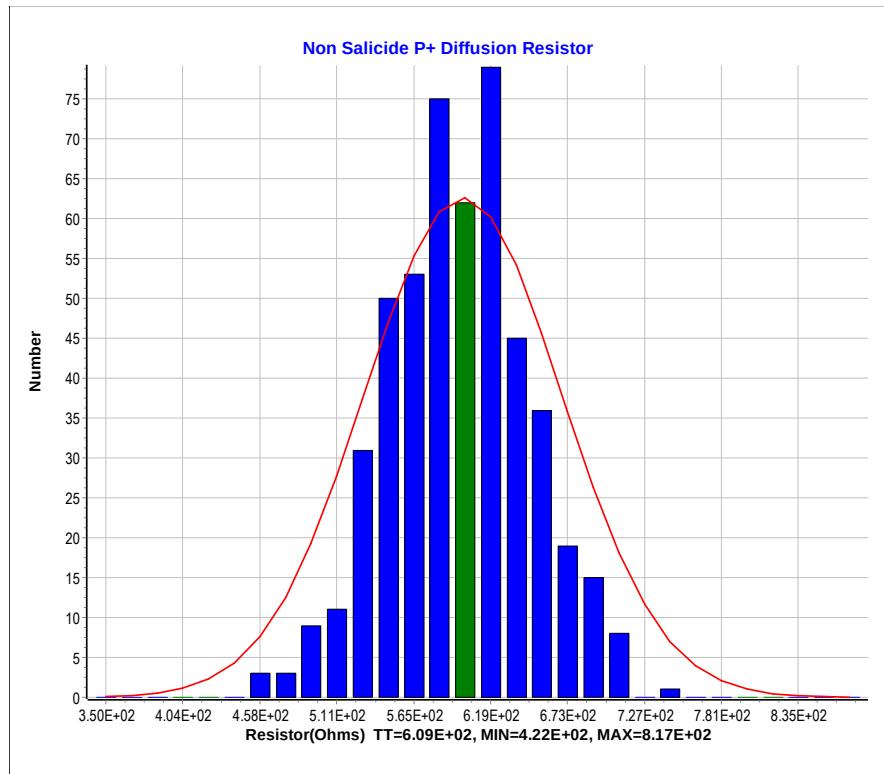


Figure 2-2 Monte Carlo plots of rnpd_l110e Resistor

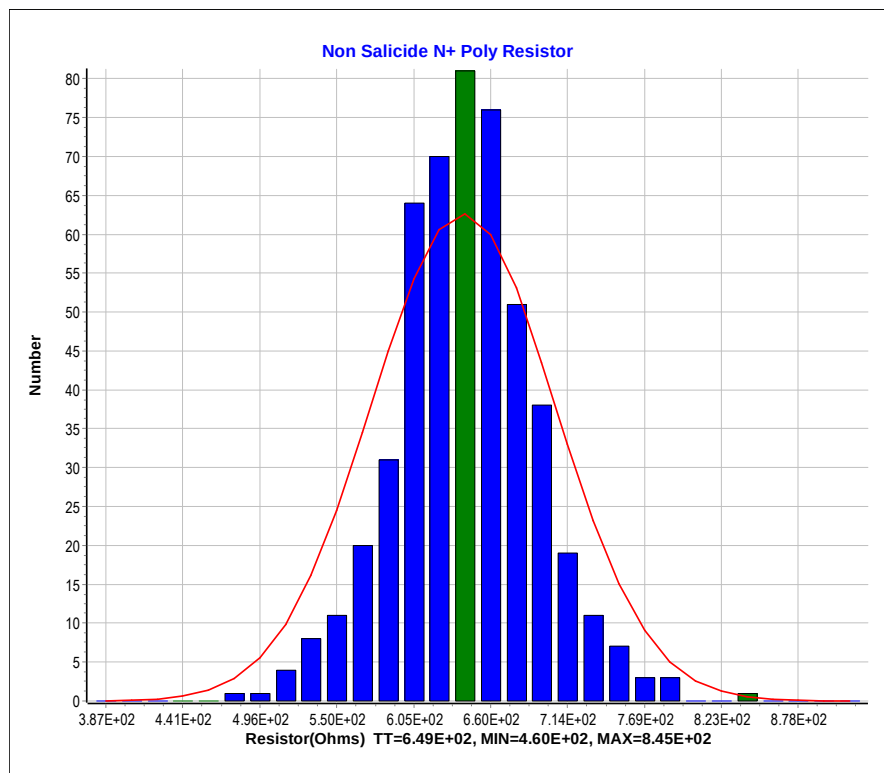


Figure 2-3 Monte Carlo plots of rnnpo_l110e Resistor

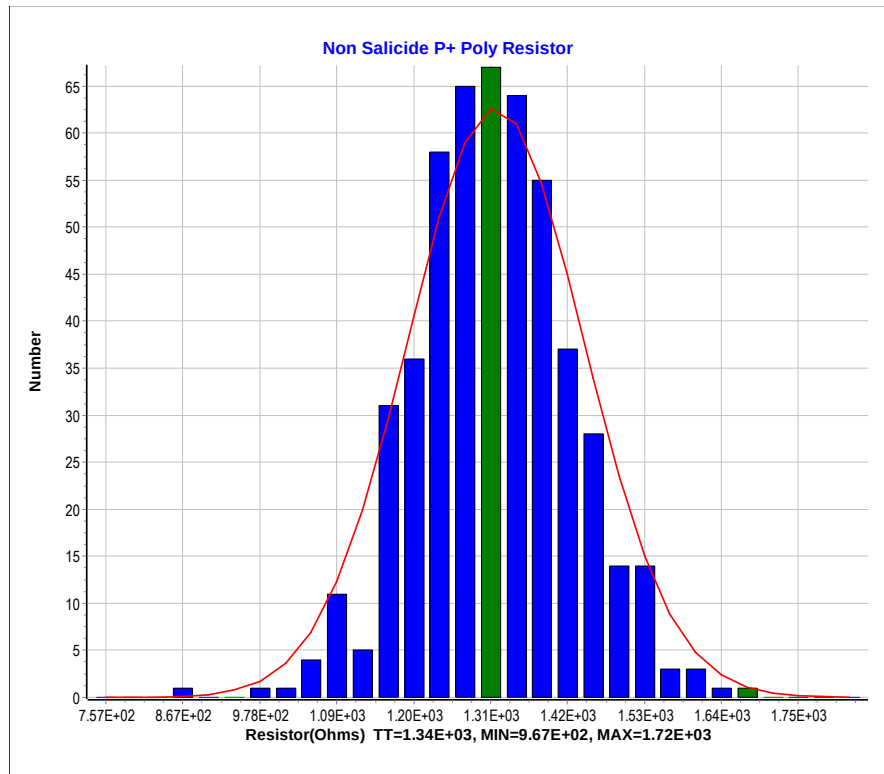


Figure 2-4 Monte Carlo plots of rnppo_l110e Resistor

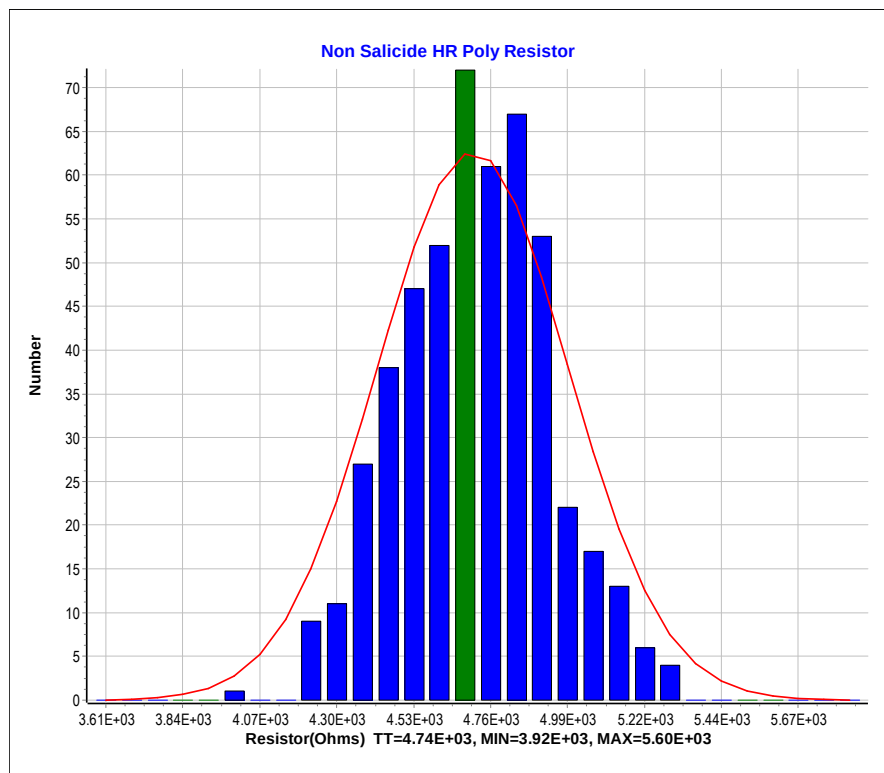


Figure 2-5 Monte Carlo plots of rnhr1000_l110e Resistor

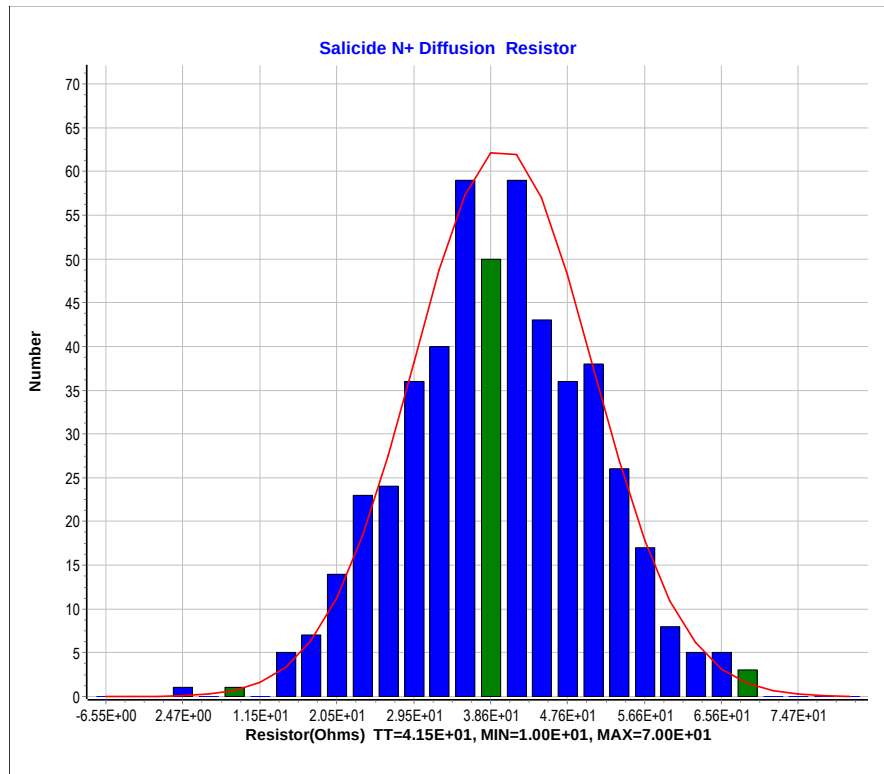


Figure 2-6 Monte Carlo plots of rsnd_mml110e Resistor

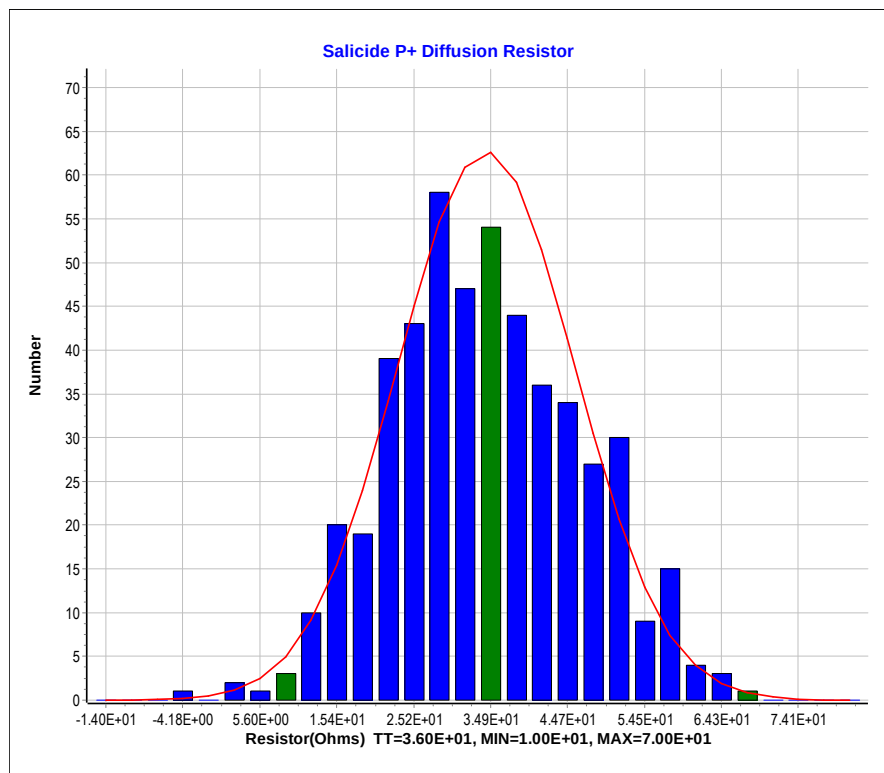


Figure 2-7 Monte Carlo plots of rspd_mml110e Resistor

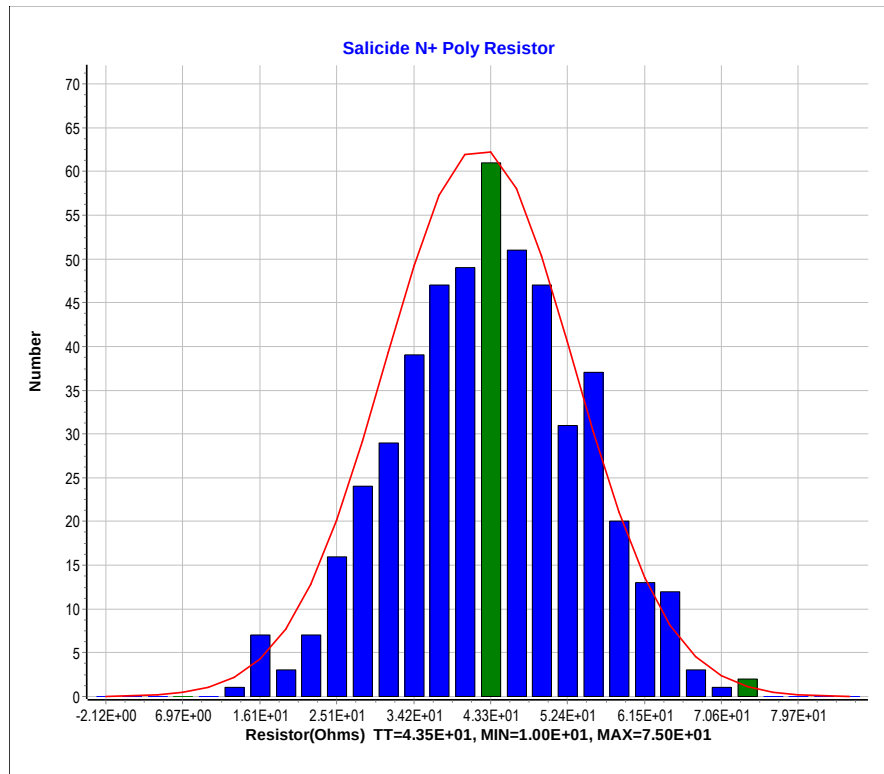


Figure 2-8 Monte Carlo plots of rsnp0_mml110e Resistor

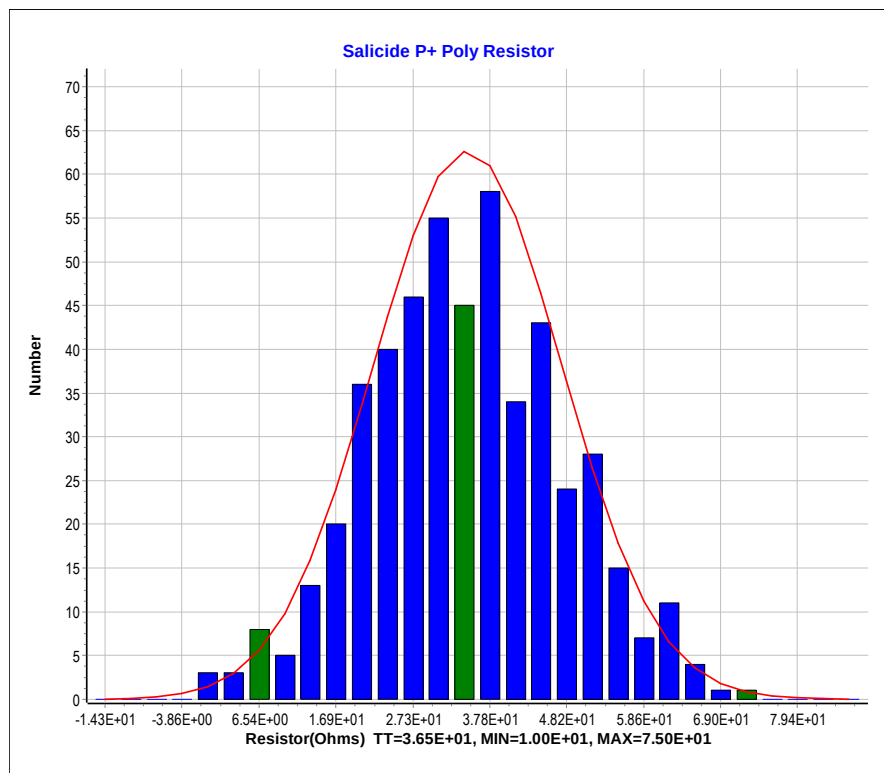


Figure 2-9 Monte Carlo plots of rspp0_mml110e Resistor

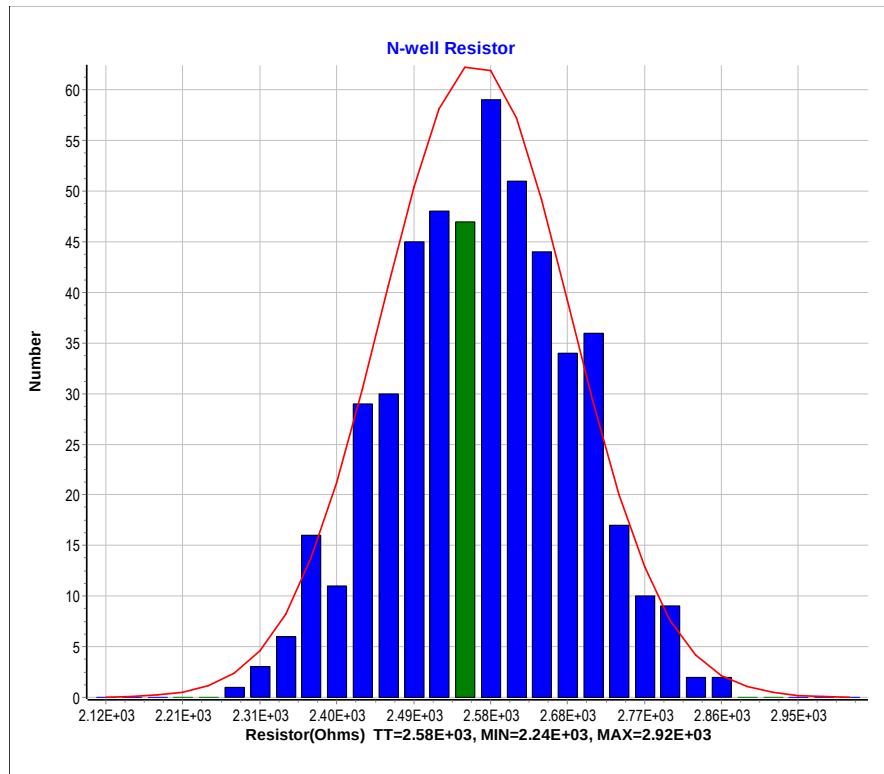


Figure 2-10 Monte Carlo plots of rnwell_mml110e Resistor(Before shrink:W=2um,L=10um)

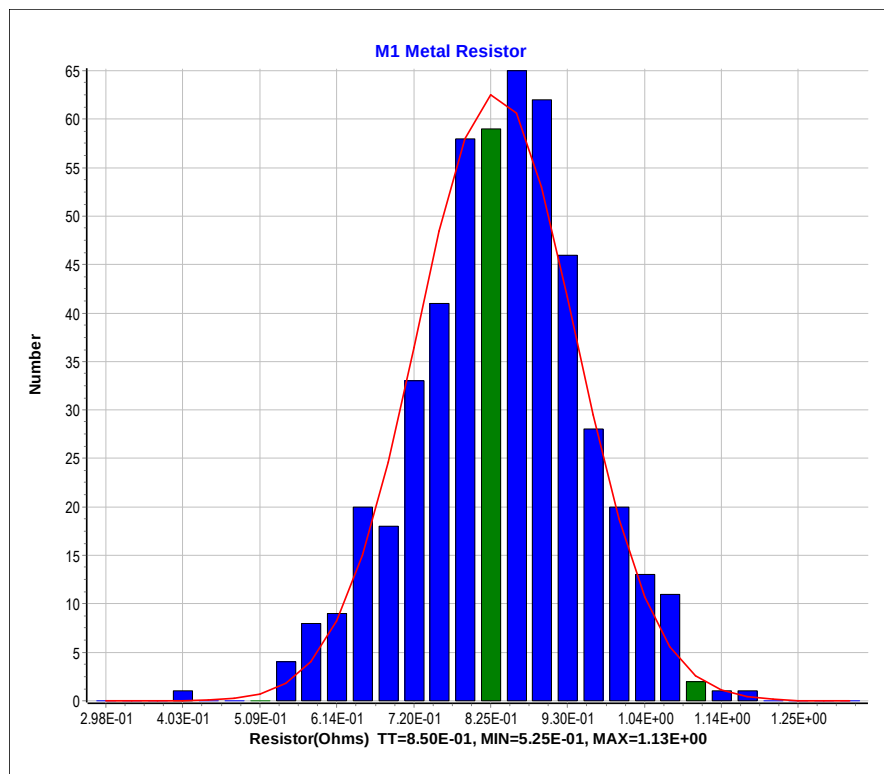


Figure 2-11 Monte Carlo plots of r_m1_l110e Resistor(Before shrink:W=1um,L=5um)

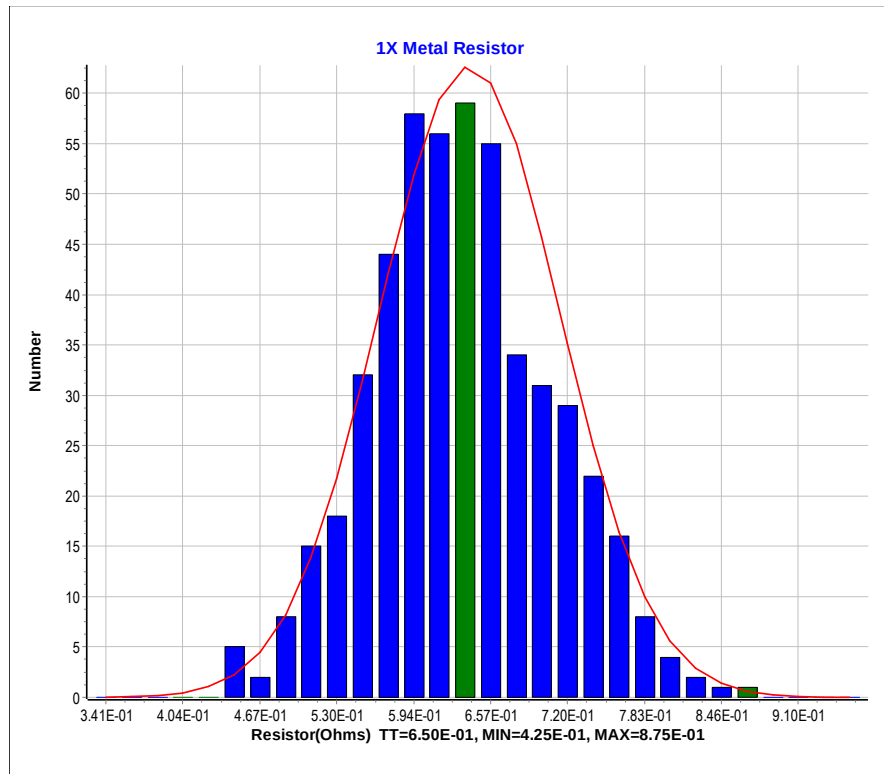


Figure 2-12 Monte Carlo plots of r_1x_l110e Resistor(Before shrink:W=1um,L=5um)

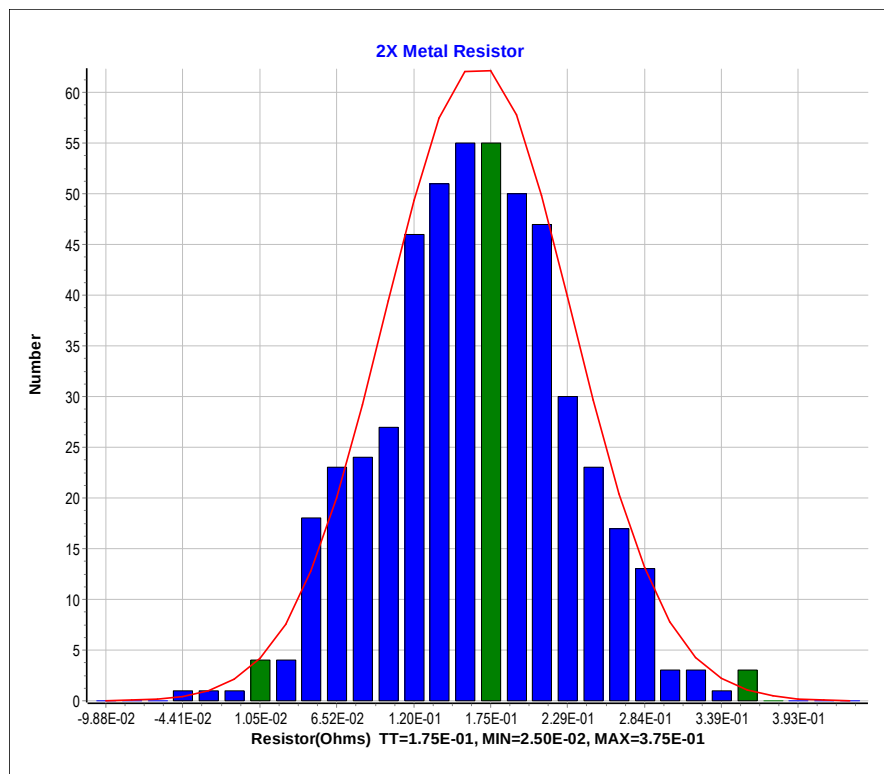


Figure 2-13 Monte Carlo plots of r_2x_l110e Resistor(Before shrink:W=1um,L=5um)

3 Monte Carlo Simulation Results for Mismatch

3.1 Simulation Result Plots

The following figures show the Monte Carlo mis-matching simulation results of Resistor at 25 °C.

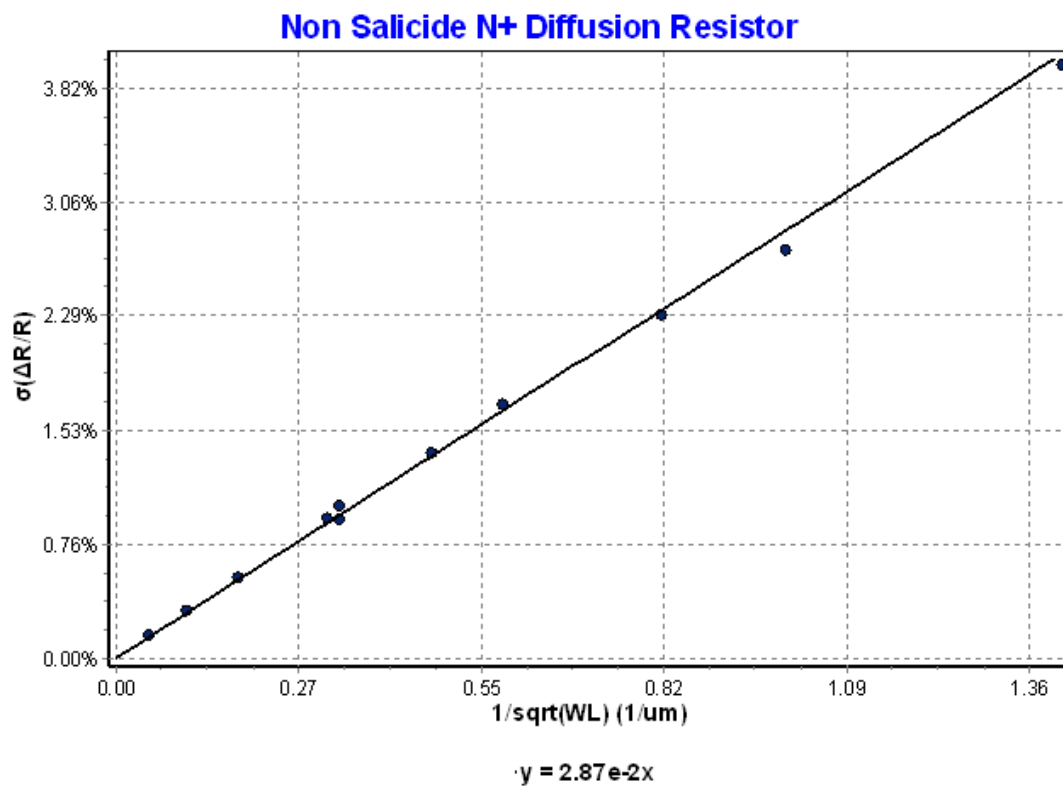


Figure 3-1 Mis-matching simulation of rnnd_l110e Resistor

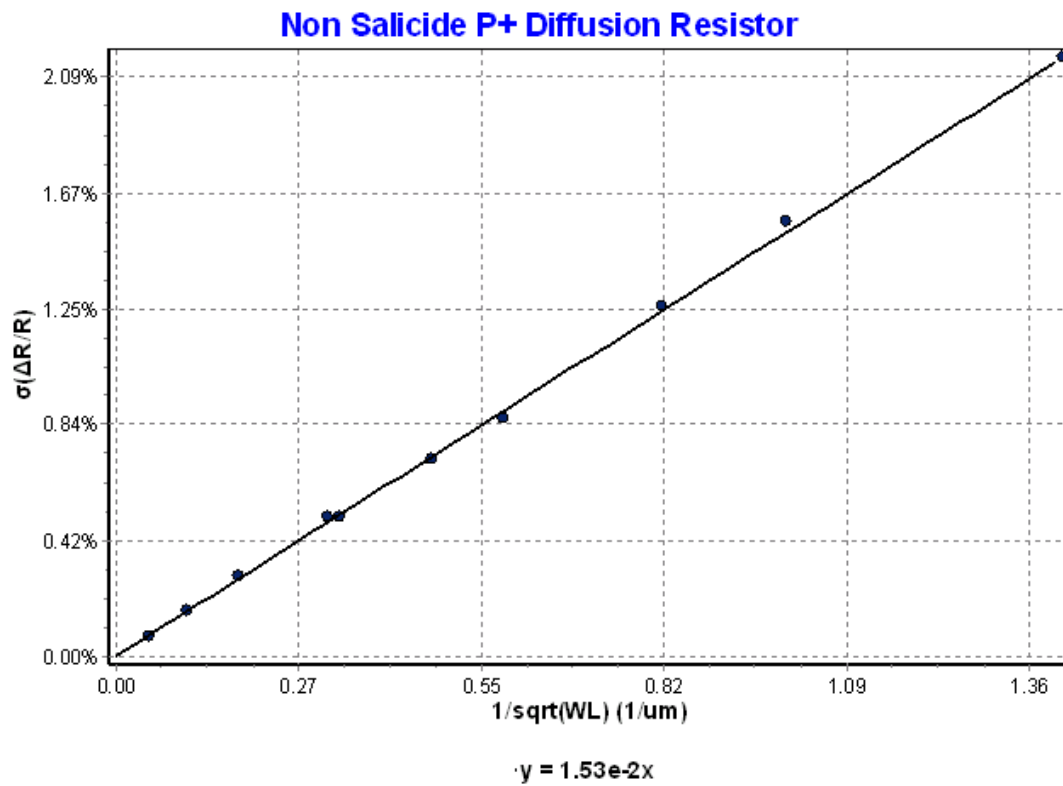


Figure 3-2 Mis-matching simulation of rn timer_l110e Resistor

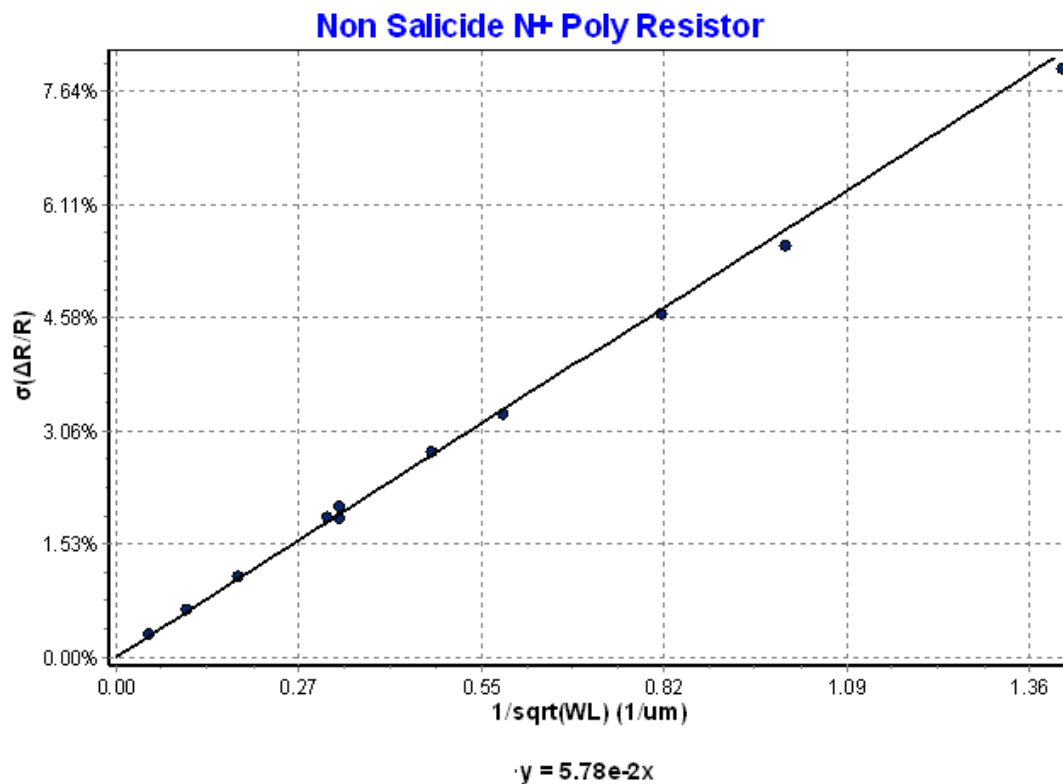


Figure 3-3 Mis-matching simulation of rnnpo_l110e resistor

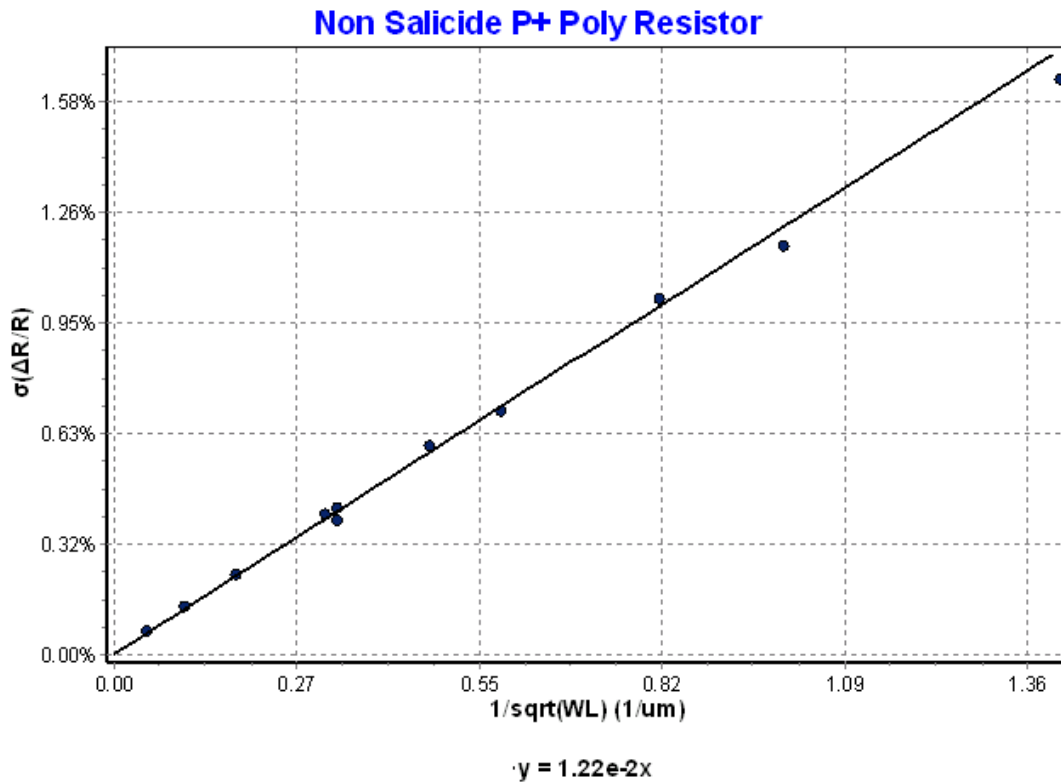


Figure 3-4 Mis-matching simulation of rnppo_l110e Resistor

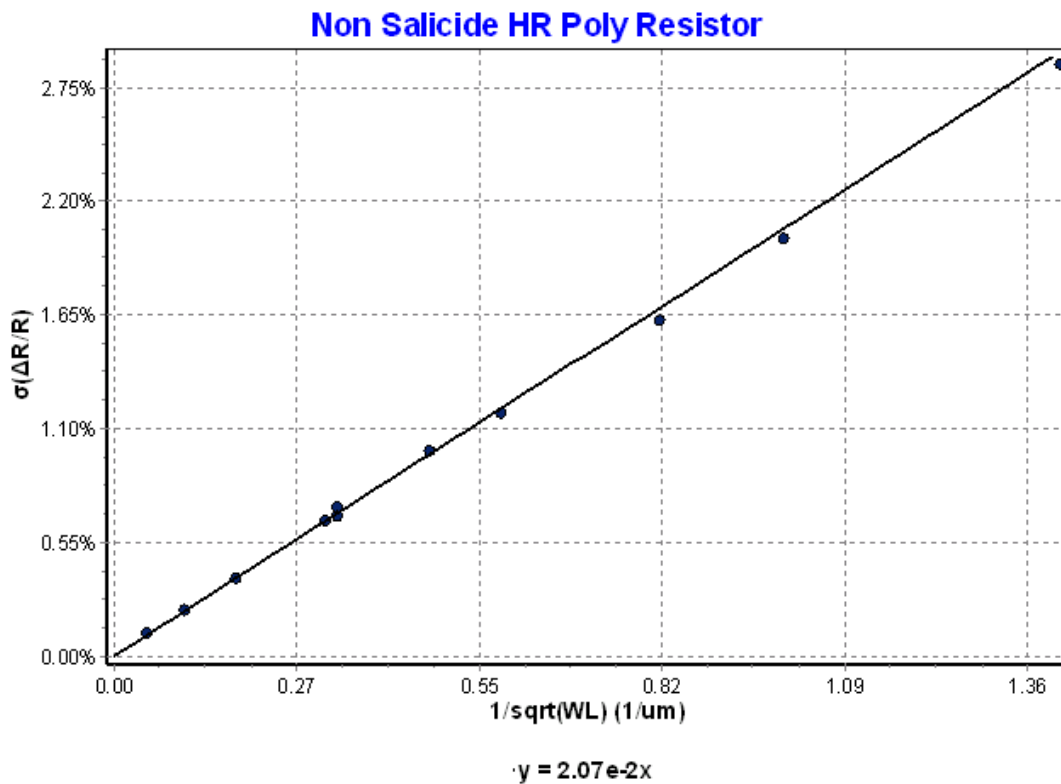


Figure 3-5 Mis-matching simulation of rnhr1000_l110e resistor

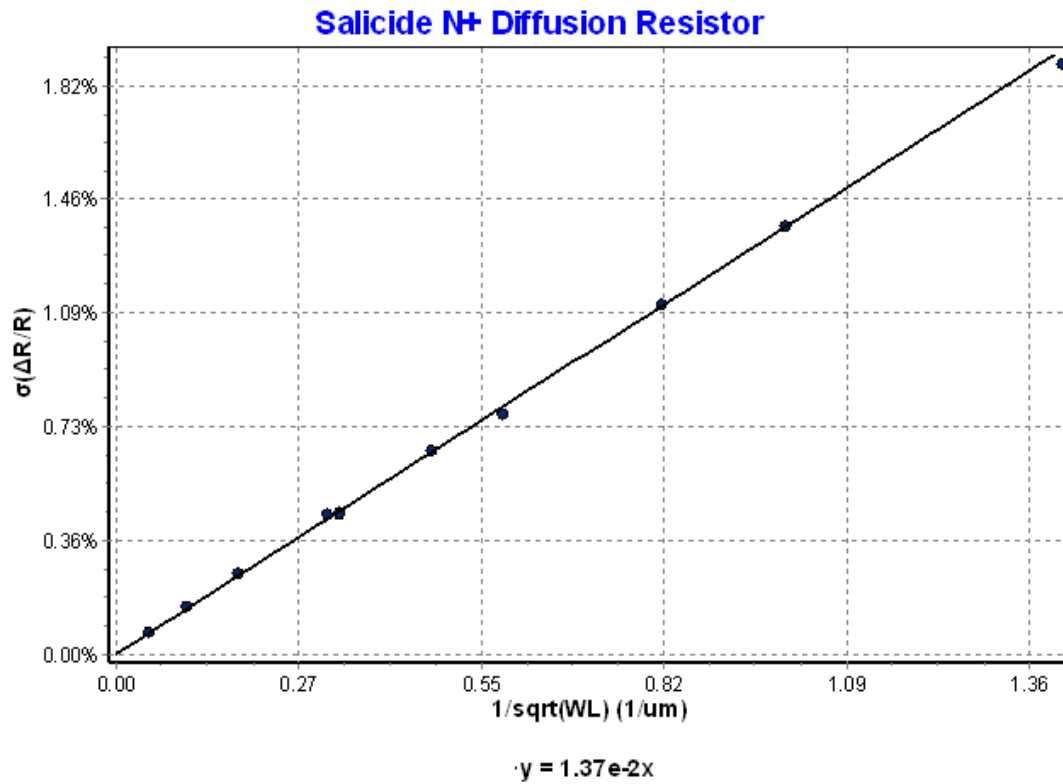


Figure 3-6 Mis-matching simulation of rsnd_mml110e Resistor

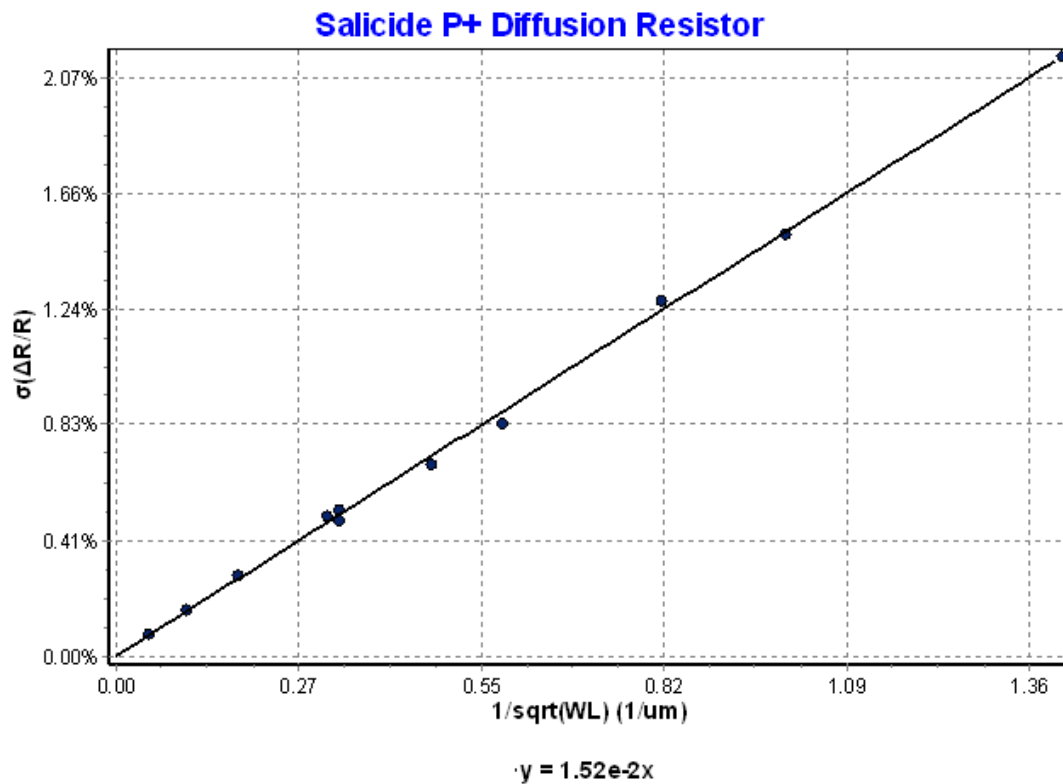


Figure 3-7 Mis-matching simulation of rspd_mml110e resistor

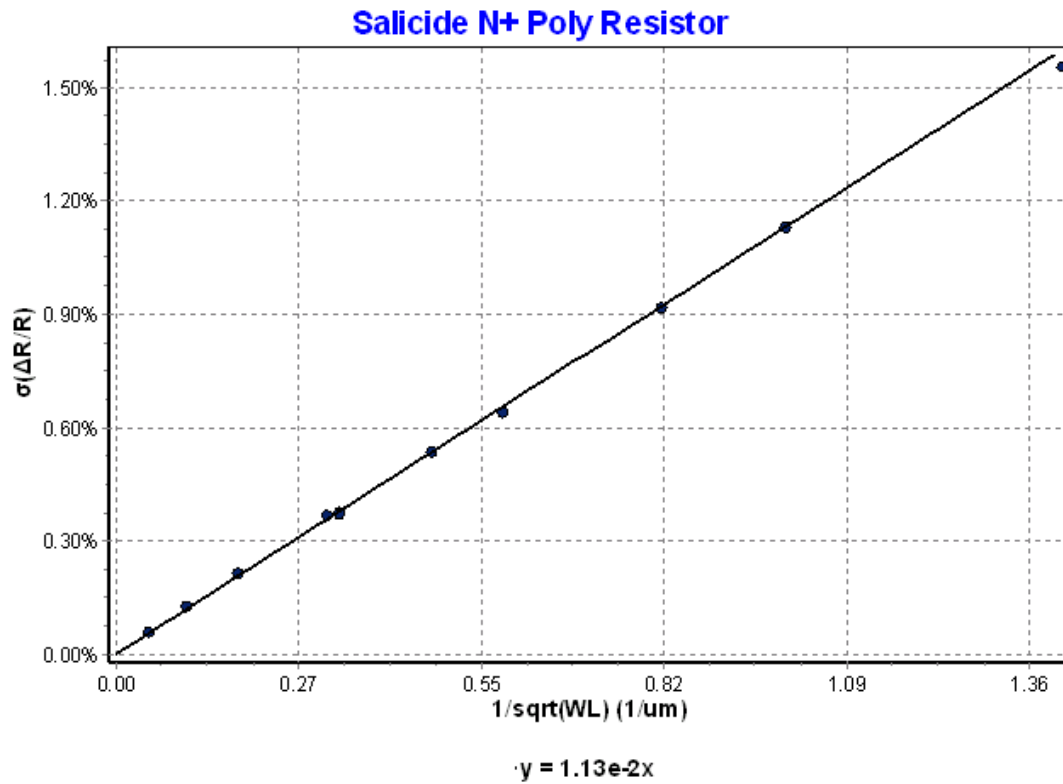


Figure 3-8 Mis-matching simulation of rsnpo_mml110e Resistor

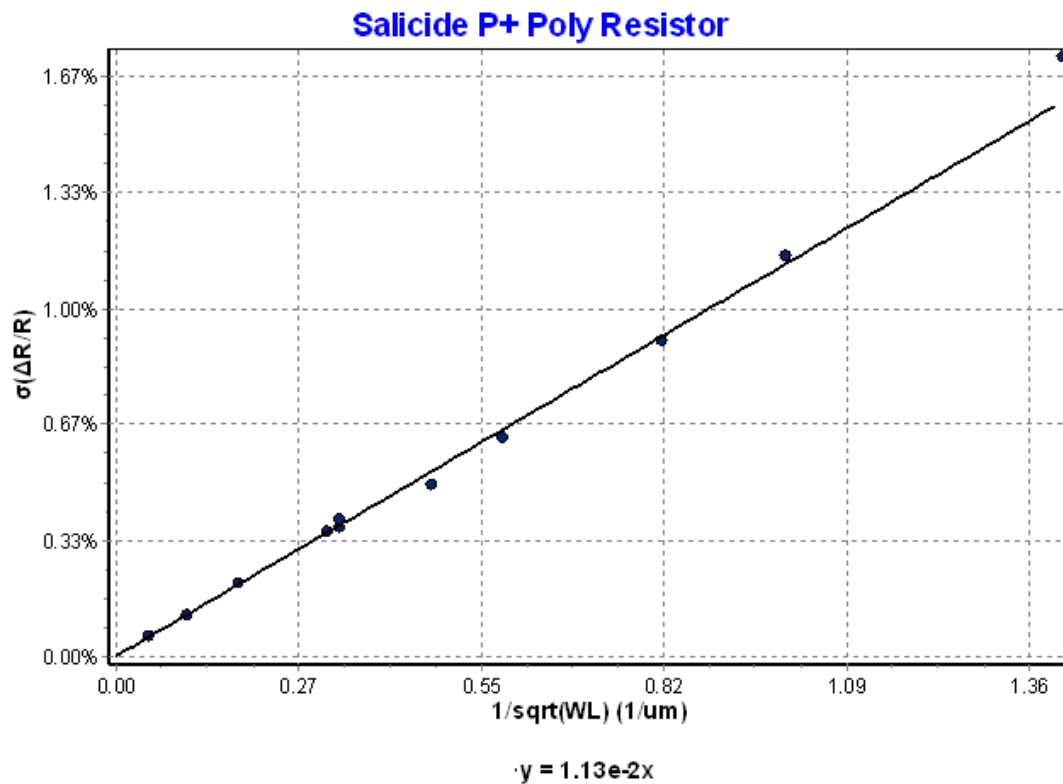


Figure 3-9 Mis-matching simulation of rsppo_mml110e resistor

4 Appendix

4.1 HSPICE Warning Message

Table 4-1 Hspice Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.2 ELDO Warning Message

Table 4-2 Eldo Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.3 SPECTRE Warning Message

Table 4-3 Spectre Warning Message

NO.	Warning Message	Explanation
1.	None	None